Notifications

# 2. Plotting a time series

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Calculator

Recitation due Mar 6, 2024 20:30 IST

# Creating uniformly spaced vectors

Start of transcript. Skip to the end.

a common step in many programming tasks

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a common step in many programming tasks is to create a list of values between an

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0:00 / 0:00 2.0x

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## LTI Consumer (External resource) (1.0 points possible)

MATLAB Documentation (https://www.mathworks.com/help/)

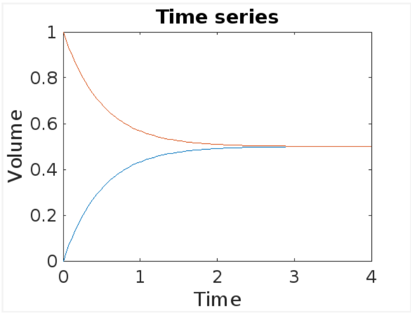
```
%%% Copy and paste all of the correct code from the previous part here
% Input the 2x2 matrix describing your linear system as a variable A
A = [-1 1; 1 -1];
% Input your initial conditions as a column vector x0
x0 = [0; 1];
% Use eig(A) to find the eigenvalues and eigenvectors of A.
% Then define the eigenvectors as column vectors v1 and v2, and the eigenval
% so that A*v1 = lambda1*v1, etc.
[V,D] = eig(A);
v1 = V(:,1);
v2 = V(:,2);
lambda1 = D(1,1);
lambda2 = D(2,2);
% Calculate the column vector c = [c1;c2] from the initial conditions using
c = inv(V)*x0;
c1 = c(1,1); c2 = c(2,1);
%%% Define a row vector t with 100 equally spaced entries,
%%% beginning with 0 and ending at 4.
t = linspace(0,4,100);
%%% Define two row vectors h1 and h2, with entries corresponding
%%% to h1(t) and h2(t) evaluated at each time in t.
h1 = c1*exp(lambda1*t)*v1(1) + c2*exp(lambda2*t)*v2(1);
h2 = c1*exp(lambda1*t)*v1(2) + c2*exp(lambda2*t)*v2(2);
%%% Now use plot to plot the two vectors against time on the same figure
%
plot(t,h1) %tell MATLAB to plot h1 here
%
hold on % You need to put this command in between the plot commands so that
%
plot(t,h2) %tell MATLAB to plot h2 here
%
% Don't worry about the following commands, they are just formatting commanc
set(gca,'fontsize',18)
xlabel('Time')
ylabel('Volume')
title('Time series')

```

Calculator

▶ Run Script

## Output




## 2. Plotting a time series

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